WORKED SOLUTION

STEP III 2009, Question 4

Pure • Laplace Transforms

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q4

The key idea

Parts (i)–(iii) are each one line of calculus applied to the defining integral: (i) just merges two exponentials, (ii) is a substitution and (iii) is integration by parts. Part (iv) can be done by integrating by parts twice, but it is neater to use (iii) twice and let the transforms of sin and cos pin each other down. For the final part, think of the target $\mathrm{e}^{-pt}\cos qt$ as built in three steps, $\sin t \to \cos t \to \cos qt \to \mathrm{e}^{-pt}\cos qt$, each step being one of the four results.

Solution

Write $\mathcal{L}[g](s)$ for the Laplace transform of a function g, so that $F = \mathcal{L}[f]$. "Suitable" means the integrals converge; in particular we assume $\mathrm{e}^{-st}f(t) \to 0$ as $t \to \infty$ for $s > 0$.

Part (i)

Combining the exponentials,

$$\mathcal{L}[\mathrm{e}^{-bt}f(t)](s) = \int_0^\infty \mathrm{e}^{-st}\mathrm{e}^{-bt}f(t)\,\mathrm{d}t = \int_0^\infty \mathrm{e}^{-(s+b)t}f(t)\,\mathrm{d}t = F(s+b),$$

which is the definition of F evaluated at $s+b$. This is legitimate because $s+b > 0$.

Part (ii)

Substitute $u = at$, so $t = u/a$ and $\mathrm{d}t = \mathrm{d}u/a$. Since $a > 0$ the limits 0 and ∞ stay as they are, so

$$\mathcal{L}[f(at)](s) = \int_0^\infty \mathrm{e}^{-st}f(at)\,\mathrm{d}t = \frac{1}{a}\int_0^\infty \mathrm{e}^{-(s/a)u}f(u)\,\mathrm{d}u = \frac{1}{a}F\left(\frac{s}{a}\right),$$

again using $s/a > 0$.

Part (iii)

Integrate by parts, differentiating e^{-st} and integrating $f'(t)$:

$$\begin{aligned}\mathcal{L}[f'](s) &= \Big[e^{-st}f(t)\Big]_0^\infty - \int_0^\infty (-s)e^{-st}f(t)\,dt \\ &= (0 - f(0)) + s\int_0^\infty e^{-st}f(t)\,dt \\ &= sF(s) - f(0),\end{aligned}$$

where the upper limit vanishes because $e^{-st}f(t) \to 0$ as $t \to \infty$ (part of f being suitable).

Part (iv)

Let $S(s) = \mathcal{L}[\sin t](s)$ and $C(s) = \mathcal{L}[\cos t](s)$. Both integrals converge for $s > 0$, because $|e^{-st}\sin t| \le e^{-st}$, and similarly for cos; and $e^{-st}\sin t$ and $e^{-st}\cos t$ both tend to 0, so (iii) applies to each.

Applying (iii) with $f = \sin$, whose derivative is cos and with $\sin 0 = 0$:

$$C(s) = sS(s).$$

Applying (iii) with $f = \cos$, whose derivative is $-\sin$ and with $\cos 0 = 1$:

$$-S(s) = sC(s) - 1.$$

(The transform of $-\sin t$ is $-S(s)$, since a constant factor comes straight out of the integral.) Substituting the first equation into the second,

$$-S(s) = s^2S(s) - 1 \quad\Longrightarrow\quad (s^2+1)S(s) = 1 \quad\Longrightarrow\quad S(s) = \frac{1}{s^2+1},$$

as required. As a by-product,

$$C(s) = \frac{s}{s^2+1}.$$

The final part

? **Ask yourself** How do we turn $\sin t$ into $\cos qt$ without any new integration?

Differentiation turns sin into cos (that's (iii)), rescaling t turns $\cos t$ into $\cos qt$ (that's (ii)), and multiplying by an exponential is (i). So we go in three steps.

Step 1: $\cos t$. By (iii) with $f(t) = \sin t$, using (iv) for F,

$$\mathcal{L}[\cos t](s) = s\cdot\frac{1}{s^2+1} - \sin 0 = \frac{s}{s^2+1}.$$

(This is the by-product of part (iv).)

Step 2: $\cos qt$. By (ii) with $f(t) = \cos t$ and $a = q > 0$,

$$\mathcal{L}[\cos qt](s) = \frac{1}{q}\cdot\frac{s/q}{(s/q)^2+1} = \frac{1}{q}\cdot\frac{sq}{s^2+q^2} = \frac{s}{s^2+q^2}.$$

Step 3: $\mathrm{e}^{-pt}\cos qt$. By (i) with $f(t)=\cos qt$ and $b=p>0$, we replace s by $s+p$:

$$\boxed{\mathcal{L}[\mathrm{e}^{-pt}\cos qt](s)=\frac{s+p}{(s+p)^2+q^2}} \qquad (s>0).$$

Checks

- Putting $p\to 0$ and $q=1$ gives $s/(s^2+1)$, the transform of $\cos t$ from Step 1. Putting $q\to 0$ gives $1/(s+p)$, and indeed $\int_0^\infty \mathrm{e}^{-(s+p)t}\,\mathrm{d}t = 1/(s+p)$.
- As $s\to\infty$, $sF(s)\to 1=\cos 0$, as it must: for large s the weight $s\mathrm{e}^{-st}$ concentrates near $t=0$, so $sF(s)$ tends to the value of the function at 0.
- Numerically, with $s=1$, $p=\frac{1}{2}$, $q=2$ the formula gives $1.5/6.25=0.24$, and Simpson's rule on the integral agrees. The computer checked (i)–(iv) and the final answer at random parameter values to within 10^{-12}.

Common slips

- In (ii), forgetting the factor $1/a$ from $\mathrm{d}t=\mathrm{d}u/a$, or writing $F(as)$ instead of $F(s/a)$.
- In Step 3, shifting only some of the s's: *every* s becomes $s+p$, so the numerator is $s+p$, not s.
- Getting the sign of $f(0)$ wrong in (iii): the boundary term is $0-f(0)$.

Beyond STEP

These rules are the engine of the Laplace transform method for differential equations: (iii) turns derivatives into multiplication by s, so a linear ODE with constant coefficients becomes an algebraic equation for $F(s)$, which you solve and then "look up" in a table built from results like the one here. The shift rule (i) is exactly why damped oscillations $\mathrm{e}^{-pt}\cos qt$ show up as poles at $s=-p\pm \mathrm{i}q$. Boyce and DiPrima, *Elementary Differential Equations*, has a full chapter on the method.

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